



第一次作业

数据库第一次作业

后续题目涉及员工 `ID` 属性，但教材原图里没有。这里假设employee和works表均具有该属性：

- `employee(person_name, street, city, ID)`
- `works(person_name, company_name, salary, ID)`
- `company(company_name, city)`

2.6

a.

$$\Pi_{person_name}(\sigma_{city="Miami"}(employee))$$

b.

$$\Pi_{person_name}(\sigma_{salary>100000}(works))$$

c.

$$\Pi_{person_name}(\sigma_{city="Miami" \wedge salary>100,000}(employee \bowtie works))$$

2.8

a

$$\Pi_{person_name, ID}(works) - \Pi_{person_name, ID}(\sigma_{company_name="BigBank"}(works))$$

b.

$$\Pi_{ID, person_name}(works) - \Pi_{W2.ID, W2.person_name} \left(\sigma_{W1.salary > W2.salary} (\rho_{W1}(works) \times \rho_{W2}(works)) \right)$$

2.14

a.

$$\Pi_{ID, person_name} (\sigma_{company_name = \text{"BigBank"}}(works))$$

b.

$$\Pi_{ID, person_name, city} (\sigma_{company_name = \text{"BigBank"}}(works \bowtie employee))$$

c.

$$\Pi_{ID, person_name, street, city} (\sigma_{company_name = \text{"BigBank"} \wedge salary > 10000}(works \bowtie employee))$$

d.

$$\Pi_{ID, person_name} ((employee \bowtie works) \bowtie company)$$